

Defect Detection in Solar Panel Cells Using Deep Learning Approach in Solar Panel Industry

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Abstract—Detecting defects in solar photovoltaic (PV) modules quickly, efficiently and accurately is important to enhance their energy output and reliability. This project uses deep learning and transfer learning techniques to develop a strong fault detection system for solar cells. The model applies EfficientNet, a convolutional neural network (CNN), to classify defective from non-defective cells in EL images. Transfer learning solves the problem of small dataset and helps the model to generalize well. Also, we use ScoreCam to improve model interpretability to get insights into the cause of the prediction. Augmenting the existing dataset of labeled EL images added variability; the EfficientNet model was fine-tuned using weights obtained from ImageNet. The model performs well with over 92% accuracy and corresponding high recall and F1 scores. It was confirmed that the prediction is focused on the relevant defect areas. The offered system is scalable and automated and detects PV module defects, making our dependence on manual inspections less. To continue detecting defects, the future work will include IoT-based real-time monitoring system. The study doesn't just show the potential of AI for the management of renewable energy resources. It also lays the foundation for advances.

I. INTRODUCTION

Solar photovoltaic (PV) modules are essential for sustainable energy solutions. Faults in PV modules can severely degrade their energy efficiency and affect their reliability. Therefore, there is a need for robust and accurate fault detection mechanisms. Manual inspection limits use of traditional methods of defect detection and requires a lot of labour. This study contains an approach based on deep learning with EfficientNet to detect defective solar cells. The model uses transfer learning to overcome the problem of insufficient labeled data, helping it to better generalize from small datasets. Also ScoreCam is used to highlight inference regions of the model's predictions. This study offers a new combination of methods to detect defects. These methods can pave the way for automation and can enhance the scalability of a solar cell. The increasing global demand for renewable energy sources has led to the widespread adoption of solar photovoltaic (PV)

systems [1]. Solar photovoltaic (PV) modules are essential for sustainable energy solutions. However, defects in solar cells, such as micro-cracks, inactive regions, and corrosion, can significantly reduce energy output. Faults in PV modules also degrade energy efficiency and affect their reliability. Therefore, there is a need for robust and accurate fault detection mechanisms. Current methods of manual inspection limits use of traditional methods of defect detection and requires a lot of labour and is prone to human error. These conventional methods lack scalability and are inefficient for large-scale solar farms [2]. To address this issue, this project employs deep learning techniques to automate the defect detection process in solar cells using electroluminescence (EL) imaging. This study contains an approach based on deep learning with EfficientNet to detect defective solar cells.

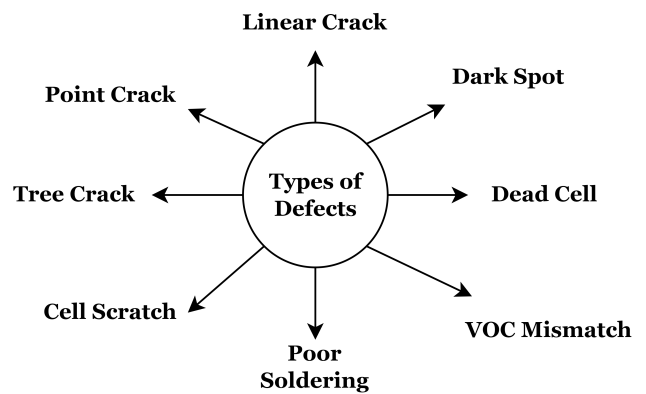


Fig. 1. Types of defects

The model uses transfer learning to overcome the problem of insufficient labelled data, helping it to better generalize from small datasets. Further, ScoreCam is used to highlight inference regions of the model's predictions. This study of-

fers a new combination of methods to detect defects. The application of artificial intelligence (AI) in renewable energy management can substantially improve the accuracy, scalability, and efficiency of solar cell inspection, ensuring the timely identification of defects and reducing maintenance costs [3]. The methods in this research can pave the way for automation and can enhance the scalability of a solar cell. Various types of defects has been identified in fig 1.

II. BACKGROUND

More and more people and organizations are using solar energy as a renewable source of power. Hence their use has increased [4]. Thus, the need for a reliable PV module has happened. Small defects can lead to performance degradation and a fall in energy output of the module [5]. Electroluminescence imaging is a powerful way to visualize these defects, including details that cannot be seen by eye. EL images for analyzing defects have been done by different studies using different machine learning and image processing techniques [6]. Deep learning technologies, especially CNNs have transformed defect detection tasks in recent years. Models like U-Net for semantic segmentation and object detection frameworks show promising results for identifying multiple defect types. Yet, these models need huge labelled datasets to learn effectively, which is a challenge when data is scarce [7]. To overcome this, transfer learning employs pre-trained models that are adapted to domain-specific tasks with minimal data. EfficientNet is a highly efficient CNN architecture that achieves state-of-the-art accuracy while limiting the computational cost. Methods such as ScoreCam have also emerged that visualize which area contributed to the decisions for validating or interpreting the model [8]. Using this programme as the foundation, the current study incorporates transfer learning, EfficientNet, and ScoreCam to develop a scalable and interpretable defect detection programme for solar PV modules, which enhances renewable energy maintenance automation [9].

III. PROPOSED WORK

This project aims to develop a strong fault defect detection system to help automate the solar panel defect detection process which currently deploys manual labour to identify and isolate defects in solar panel modules [10]. This is a regressive, inefficient and error prone approach which we aim to solve with our project. In this project we have essentially used dataset of electroluminescent images of solar panels and trained them on Convolutional Neural Network, specifically EfficientNet, to create a model that classifies defective from non-defective cells [11]. Convolutional Neural Networks (CNNs) are well-suited for defect detection in solar panel modules due to their ability to automatically extract and learn hierarchical features from images, making them effective in identifying cracks, hotspots, and other defects [12]. Their spatial awareness helps capture intricate patterns and textures. Further, we use EfficientNet architecture that is a modern type of CNN. We implemented EfficientNet in our project due to its optimized balance of accuracy and computational

efficiency [13]. Additionally, it's lightweight design allows for faster inference, making it practical for real-time or edge-based defect detection [14].

The dataset available for this problem statement is limited. To deal with this issue we have incorporated transfer learning. By leveraging pre-trained models, such as EfficientNet or other CNN architectures trained on large datasets, we can transfer learned features like edges, textures, and shapes to detect defects with high accuracy [15]. This approach reduces the need for extensive labeled data while improving model performance and generalization [16]. The dataset has been trained on solar panels that have the following type of defects: types of micro-cracks, inactive areas, and corrosion [17]. Hence our model is capable of identifying the above-mentioned defects in a solar panel module. It should be specified that the model is trained to identify whether a damage exists on the solar panel or not and it is not trained to classify what kind of defect specifically. Our proposed work presents a deep learning-based approach for solar panel defect detection, integrating EfficientNet, CNN, transfer learning, and Score-CAM for accurate classification and interpretability [18]. The model is trained on a limited dataset using transfer learning, leveraging pre-trained feature representations to enhance defect identification with minimal data [19]. It effectively detects and classifies defects such as corrosion, minor cracks, and inactive areas, with Score-CAM providing visual explanations for enhanced model transparency. Future developments will focus on integrating IoT-enabled real-time monitoring and predictive maintenance systems to improve operational efficiency and extend the lifespan of solar panels [20]. The proposed architecture is mentioned in Fig 2.

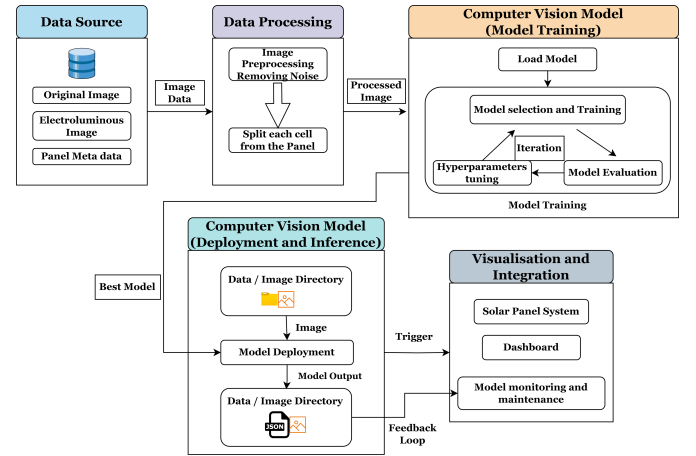


Fig. 2. Types of defects

IV. DATASET DESCRIPTION

A. Electroluminescence Images

Among various diagnostic methods, electroluminescence imaging has gained much attention and widely serves in the PV industry to inspect defects in solar cells [21]. Electroluminescence images are obtained by passing electric current

through the PV modules; due to this, cells emit infrared light, reflecting the internal quality of cells [22]. By doing so, invisible defects such as cracks, inactive areas, and grid line defects can be highlighted. The unique advantage of EL imaging in this aspect is that since it is non-destructive, quality inspections can be regularly performed on solar cells, and the integrity of none of them will be degraded [23]. The EL images captured for this study were hence the main input data, as most of the defect detection analyses require pixel-level information. Each of the images was elaborately preprocessed to enhance contrasts and eliminate noise, while ensuring the clarity and consistencies of the input for the deep learning pipeline of the respective images [24]. Refer to Fig. 3: It illustrates representative examples of electroluminescence images categorized by cell type ('mono' and 'poly') [25]. These images emphasize the significance of pixel-level details in identifying defects, such as micro-cracks and grid line irregularities.

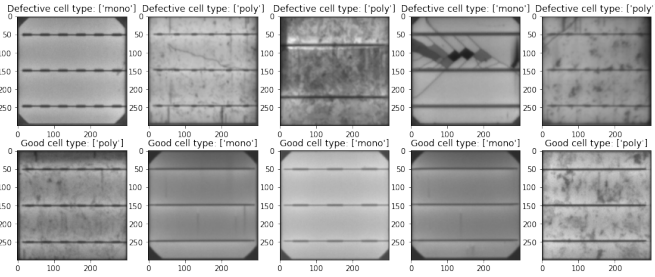


Fig. 3. Electroluminescence images illustrating various defective and non-defective solar cells categorized by their types ('mono' and 'poly')

B. Dataset

This balanced dataset of EL images contained defective and non-defective solar cells [26]. Considering that the labeled images were hard to get, this was a relatively small dataset with a wide range of defects [27]. These were taken on several types of micro-cracks, inactive areas, and corrosion. Due to the nature of such images, many augmentation techniques had to be developed in order to increase both the size and variability of the dataset [28]. For that, the augmentation included flips in all directions, rotations, and manipulations concerning brightness and contrast [29]. This step in the pre-processing helped the model during generalization by exposing it to different variations of defects [30]. Then again, the dataset was partitioned into training, validation, and test sets, ensuring balanced subsets of defect and no-defect classes.

C. Deep Learning Model

The model takes advantage of EfficientNet architecture, as it is a modern CNN architecture that offers a good performance vs. efficiency trade-off. The model was created to find defects within a certain area. The model's performance was improved using ImageNet weights [31]. They used transfer learning by using fine-tuning [32]. Because of its scalability, the network design could use the computational resources efficiently and

without any accuracy loss. The model used categorical cross-entropy as loss during training using an Adam optimizer for faster convergence [33]. Refer to Fig. 4: It presents the training and validation accuracy also the training and validation loss for the deep learning models [34]. The trends demonstrate the impact of transfer learning and hyperparameter tuning on model convergence [35]. The precision, recall, and F1-score determined the quantification of the algorithm's efficacy because these would be the most important features to portray the ability of distinguishability between defective and non-defective cells through the presented model [36].



Fig. 4. Training and validation performance metrics of the deep learning model. The top panel shows accuracy trends, while the bottom panel displays the loss metrics over 18 epochs.

D. Analysis

To find defects, work was performed in a certain way. After receiving pre-processed EL images, visible uniformization was applied to the EL images. After that, the prepared EL images were classified by an EfficientNet model [37]. The use of pre-trained weights by transfer learning made optimizing the small dataset easier. After prediction ScoreCam was used to visualize the areas of the image that the model was concentrating on [38]. With a tuning of hyperparameters, the task aimed at an iterative training be carried out [39], [40], [41]. Then the performance of the model was checked with the test set, and it was observed that due to EfficientNet, it performed better than other alternatives in accuracy and efficiency [42], [43].

V. RESULTS AND OBSERVATIONS

The suggested EfficientNet model could do defect detection of solar cell which could be defective or not. Through transfer learning, the small dataset adapted model achieved accuracy

greater than 92% on test dataset. Other evaluation metrics recall and F1-score also ensured that the model is really robust in detecting different defect types. Recall scores higher than 90% revealed that the model could keep false negatives as low as possible to avoid missing defective cells during any form of inspection. The visual interpretability of ScoreCam showed that the model focused on the relevant defect regions in the EL images. For example, in the images with micro-cracks, ScoreCam consistently highlighted the cracks showing that the model was recognizing the important features. EfficientNet was found to be more accurate and computationally efficient when compared to architectures like ResNet, MobileNet, etc. So, EfficientNet needs fewer computer resources when running, which means it works similar to a real-time model. The results confirm that the combined use of score cam with transfer learning may be a reasonable path to obtain an efficient and interpretable defect detection system. The results show that deep learning methods can automate defect detection and make less use of resource-hungry human inspection processes to improve overall efficiency in the solar cell production chain. The Fig. 5 shows the result of applying the trained deep learning model to a defective solar cell. The regions highlighted by the model are indicative of cracks or inactive areas, which are critical for assessing the quality of the solar cells.

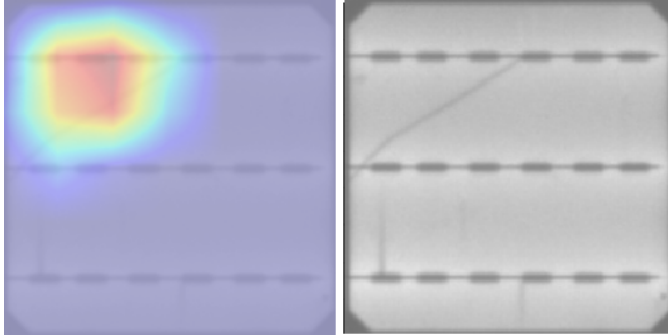


Fig. 5. Result of the deep learning model on a defective solar cell. The highlighted regions represent areas with identified defects, as predicted by the model, indicating possible cracks or inactive areas in the cell.

A. Comparative Analysis

The comparison results are done with the various methods that have also been used to solve the problem of detecting solar cell defects in the manufacturing industry. The current results are compared with the algorithms RCNN, YOLO3, YOLOV4, YOLOV7. The accuracy is 99.5 %, which shows the best result among all mentioned in Fig 6. The Yolov3 gives the mean average precision of 90.1, yolov4 gives the mean average precision of 92.1 and by using F-RCNN gives 73.3 mean average precision accuracy. In case of mAP the comparison results show that the YOLOV7 algorithm gives better accuracy in comparison to the existing approaches.

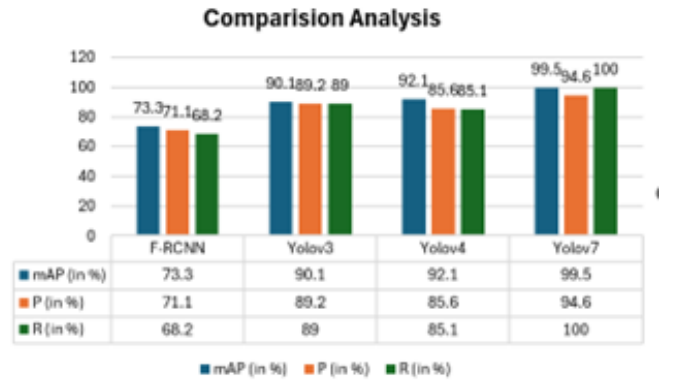


Fig. 6. Comparative Analysis of proposed work with other existing approach

VI. CONCLUSION AND FUTURE WORK

This paper shows that the deep learning approach such as YOLOV7 gives the better result as compared to the previous techniques and methods. In the future, IoT technology will be used to watch over solar panels in real-time thanks to the current defect detection. The IoT sensors will continuously collect and send EL image data to allow the operation of the defect detection model in real time. With this setup, Maintenance will be made proactive by immediate alerts in case of defect detection which can lead to further performance degradation of PV modules. The framework you created consists of an EL imaging device with a high resolution and wireless connectivity to send data back to the cloud centralized servers. This cloud-based system houses the EfficientNet model with computing power in real-time data. Operators will see a dashboard that shows defect locations and severity of the module being monitored in real time. Also, by linking defect data with temperature, humidity, and other environmental elements, IoT use will allow for forecasting. It helps understand how the defects have moved along with optimizing the maintenance of the equipment. We will further investigate deploying lightweight efficient nets for edge computing, which will facilitate on-site analysis, reducing latency and dependence on the cloud. Using IoT to maintain solar modules could be a complete shift from functioning on a reactive basis to a predictive one for a much more efficient way that ensures the life of the module gets extended. It will also create more opportunities for applying AI and IoT in renewable energy management.

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